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MTH202 (Mega File)

Mid Term (Live Quiz)

❑ Composite Relation symbolically written as:

- $SOR = \{(a,c) \mid a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$
- $SOR = \{(a,c) \mid a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$ ✓
- None of the above
- $SoR = \{(a,c) \mid a \in A, c \in C, \exists b \in B, (a,b) \in R \text{ and } (b,c) \in S\}$

❑ Subtraction (-) is a binary operation on the set of integers.

- False ✓
- True

❑ Let the set $A = \{1, 2, 3, 4\}$. Then the relation $\{(2, 4), (4, 2)\}$ is:

- Reflexive
- Transitive
- Symmetric ✓
- Antisymmetric

❑ Let $A = \{1, 2, 3, 4\}$ and define relation $R = \{(1, 1), (1, 3), (2, 4), (3, 1), (4, 2)\}$ on A. Then which of the following statement about R is true?

- R is symmetric
- R is transitive
- R is reflexive



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- R is not symmetric ✓

25 = 1(mod 3) means that 3 divides:

- 25 - 1 ✓
- 25 ° 71
- 25 + 1

The matrix representation of an inverse relation can be obtained by taking the:

- Determinant
- Adjoint
- Inverse
- Transpose ✓

Which of the following relations are functions?

- $R_1 = \{(3, 4), (4, 5), (6, 7), (8, 9)\}$
- $R_2 = \{(3, 4), (4, 5), (6, 7), (3, 9)\}$
- $R_3 = \{(-3, 4), (4, -5), (0, 0), (8, 9)\}$
- $R_4 = \{(8, 11), (3, 4), (5, 6), (17, 8), (19, 9)\}$ ✓

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The number of elements in the power set of $\emptyset$ denoted by $P(\emptyset)$ is:

- 2^0 ✓
- n
- 2^2



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7 Suppose A has 3 elements and B has 4 elements, then the number of the binary relations from A to B that are functions is:

- $2^6 / 2^8$
- $2^6 / 2^{12}$
- $2^1 / 2^6$
- $2^6 / 2^4$ ✓

8 Logic gate NOT does not define a binary operation on $\{0,1\}$ because:

- It takes a single input and gives a single output ✓
- It takes two inputs and gives a single output.
- It takes a single input and gives more than one output.
- It takes two inputs and gives multiple outputs.

2. Let R and S be reflexive relations on a set A then $R \cap S$ is reflexive.

- True ✓
- False

3. Let $R = \{(1, 2), (3, 4), (5, 6), (7, 8)\}$. Domain of the inverse of the relation is:

- $\{1, 3, 5, 7\}$
- $\{5, 6, 7, 8\}$
- $\{2, 4, 6, 8\}$ ✓
- $\{1, 2, 3, 4\}$

4. If a set A has 15 elements then $P(A)$ (power set of A) has _____ elements.

- 2^{16}



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- 2^{15}
- 2^{14}
- 2^{13} ✓

5. The matrix representation of an inverse relation can be obtained by taking the _____ of the relational matrix.

- adjoint
- Determinant
- inverse
- transpose ✓

6. Select the correct option for:

- Set of positive Real numbers
- Set of integers
- Set of negative Real numbers
- Set of Real numbers ✓

7. Range of function $f(x) = e^{Ax}$ is:

- Set of positive Real numbers ✓
- Set of integers
- Set of negative Real numbers
- Set of Real numbers

8. Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$ and let R be the "divides" relation from A to B i.e., for all $(a, b) \in A \times B$, $a R b$ iff a divides b . Then:

- $R = \{(6), (a)\}$



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- $R = \{(2, 2), (2, 6), (2, 8), (3, 6), (4, 8), (6, 8)\}$
- $R = \{(2, 6), (3, 6), (4, 8)\}$
- $R = \{(2, 2), (2, 6), (2, 8), (3, 6), (4, 8)\}$ ✓

9. Let R be a relation on a set A . R is transitive if and only if for all $a, b, c \in A$ then:

- $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$ ✓
- $(a, b) \in R$ then $(b, a) \in R$
- $(a, b) \in R$ and $(b, c) \in R$ but $(a, c) \notin R$
- $(a, b) \in R$ then $(b, a) \notin R$

10. Let $A = \{1, 2, 3\}$ and $B = \{2, 4\}$ then the number of binary relations from A to B are:

- 64 ✓
- 32
- 128
- 256

11. Which of the following is always true for the matrix representation of a symmetric relation?

- Matrix has always 1 in its diagonal entries
- Matrix has always 0 in its diagonal entries
- Matrix is singular
- Matrix is equal to its transpose ✓

1. If $A = \{1, 2, 3\}$ is a set and $R = \{(1, 2), (2, 2), (2, 1)\}$ is a relation on A , R is:

- ☐ Transitive
- ☐ None of these



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- ☒ Symmetric

2. Let $A = \{2, 3, 4\}$ and $B = \{2, 6, 8\}$ and let R be the "divides" relation from A to B . Then:

- ☐ $R = \{(2,2), (2,6), (2,8), (3,6), (4,8)\}$
- ☒ $R = \{(2,2), (2,6), (2,8), (3,6), (4,8), (6,8)\}$
- ☐ $R = \{(2,6), (3,6), (4,8)\}$
- ☐ $R = \{(3,6), (3,8)\}$

3. In the representation of an irreflexive relation using a matrix, all its diagonal entries will be:

- ☒ 0
- ☐ 1

4. The domain of a relation R from A to B is symbolically written as:

- ☒ $D(R) = \{a \text{ belongs to } A \text{ such that } (a,b) \text{ belongs to } R\}$
- ☐ $D(R) = \{a \text{ belongs to } B \text{ such that } (a,b) \text{ belongs to } R\}$

5. If $x \equiv -10 \pmod{15}$, which of the following integers are valid solutions for x ?

- ☐ 6
- ☒ -6

6. Let $A = \{1, 2, 3, 4\}$ and define the relation R on A by $R = \{(1,2), (2,3), (3,3), (3,4)\}$. Then:

- ☐ R is reflexive
- ☐ R is both reflexive and irreflexive
- ☒ R is neither reflexive nor irreflexive

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- ☐ R is irreflexive
-

7. Why is the function $f : \mathbb{Z} \rightarrow \mathbb{R}$ defined by $f(x) = \sqrt{x}$ not well defined?

- ☐ Because f is not defined for 5 and -5.
 - ☐ Because f is not defined for $z < 0$.
 - ☐ Because f is not defined for $z > 0$.
 - ☒ Because f gives imaginary values for all integers.
-

8. Operation of subtraction is a binary operation on the set of:

- ☐ Natural numbers
 - ☐ Whole numbers
 - ☒ Integers
 - ☐ Positive integers
-

9. If a relation $R = \{(1,1), (2,1), (1,2), (2,2)\}$ is given, then which of the following is not true about this relation?

- ☐ R is Symmetric.
 - ☒ R is Irreflexive.
 - ☐ R is Reflexive.
 - ☐ R is Transitive.
-

10. The logic gate NOT is a unary operation on $\{0, 1\}$:

- ☐ False
 - ☒ True
-



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1. If $f(x) = 4x - 8$, find $f(5)$

- ☐ 12
- ☒ 14
- ☐ 18
- ☐ 15

2. The inverse relation R^{-1} from B to A is defined as _____.

- ☒ $R^{-1} = \{(b,a) \in B \times A \mid (a,b) \in R\}$
- ☐ None of the above
- ☐ $R^{-1} = \{(b,a) \notin B \times A \mid (a,b) \in R\}$
- ☐ $R^{-1} = \{(b,a) \in B \times A \mid (b,a) \in R\}$

3. The matrix representation of an inverse relation can be obtained by taking the _____ of the relational matrix.

- ☒ transpose
- ☐ inverse
- ☐ determinant
- ☐ adjoint

4. Given a Set $A = \{a, b\}$ and $B = \{1, 2, 3\}$, then number of functions from A to B is....

- ☒ 9
- ☐ 12
- ☐ 6
- ☐ 18

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5. Range of the relation $\{(0,1), (3,22), (90,34)\}$ is

- ☐ $\{0,1,3,22,90,34\}$
- ☐ $\{0,3,90\}$
- ☒ $\{1, 22, 34\}$
- ☐ $\{0,13\}$

6. The number of elements in $A \times B$ are _____ if A is a set with '5' elements and B is a set with '4' elements.

- ☒ 20
- ☐ 10
- ☐ 15
- ☐ 25

7. The Range of $f: X \rightarrow Y$ is also called the image of _____ under f.

- ☒ True
- ☐ False

8. Let the set $A = \{1,2,3,4\}$ Then the relation $\{(2,4), (4,2)\}$ is

- ☐ Reflexive
- ☒ symmetric
- ☐ Transitive
- ☐ AntiSymmetric

9. If $A = \{1,2,3\}$ & $B = \{4,5,6\}$ and $R = \{(1,4), (2,5), (3,6), (3,4)\}$ The complementary relation is



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- ☐ $A \times B$ (intersection) R
- ☒ $A \times B$ ✓
- ☐ $A \times B$ (Union) R
- ☒ $A \times B$ (difference or $-$) R ✓

10. Range of a relation symbolically written as

- ☒ $\text{Ran}(R) = \{b \in B \mid (a,b) \in R\}$ ✓
- ☐ None of the above
- ☐ $\text{Ran}(R) = \{a \in A \mid (a,b) \in R\}$
- ☐ $\text{Ran}(R) = \{b \in B \mid (a,b) \in R\}$

Range of function $f(x) = |x|$ will be:

- ☐ Set of negative real numbers
- ☐ Set of whole numbers
- ☒ Set of positive real numbers ✓
- ☐ Set of real numbers

Let $A = \{1,2,3,\dots,50\}$ and $B = \{2,4,6,8,10\}$. Then the Cartesian product of A and B has:

- ☒ 100 ✓
- ☐ 50
- ☐ 250

Subtraction $(-)$ is a binary operation on the set of integers:

- ☐ False
- ☒ True ✓

Let $A = \{1,2,3,4,5,6,7,8,9\}$, then which of the following sets represent the partition of the set A ?

- ☒ $A = \{1,3,5,7,9\}$, $B = \{2,4,6\}$, $C = \{8\}$ ✓



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- $A=\{13,5,7,9\}$, $B=\{2,3,4,6\}$, $C=\{8\}$
- $A=\{13,5,7,9\}$, $B=\{2,4,6\}$, $C=\{8,9\}$

☐ Let $\$ = R$ and define the 'square' relation $R = \{(x,y) | z^2 = y\}$. The square relation is an:

- Not equivalence relation
- Equivalence relation ✓

☐ Suppose A has 3 elements and B has 4 elements, then the number of functions from A to B are:

- 81 ✓
- 12
- 8

☐ Let $P(A)$ be the power set of a non-empty set A and by using the definition of "subset" relation, c, for all $X, Y \in P(A)$, $X \subset Y$ if $x \in X$ then $x \in Y$. Then:

- c is a partial order relation. ✓
- c is not a partial order relation.

☐ The range of a function is alwaysthe co-domain of f:

- a superset of
- equal to
- a subset of ✓
- not equal to

☐ The relation $R = \{(12, 14), (13, 5), (-2, 7), (x, 13)\}$ is a function if x is not equal to:

- 14, 5 or 7. ✓
- 12, 13 or -2.
- 12, 13 or 2.
- 14, 7 or 13.

☐ Which of the following logical connectives is not a binary operation?

- Disjunction

All Subjects WhatsApp Groups Available

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Note: Live quiz MCQs also repeated in Exam (6, 7)



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- Negation ✓
- Implication
- Conjunction

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